Investigating the Efficacy of Simulation Coding Exercises for Teaching Probability Theory

Church, Marguerite
machurch@ucsd.edu

Ling, Rihui
riling@ucsd.edu

Ni, Jingjun
jin010@ucsd.edu

Tallada, Alyssa
atallada@ucsd.edu

Zhang, Yunqi
yuz227@ucsd.edu

Chi, Peter
pbchi@ucsd.edu

Abstract

This study investigates whether simulation-based coding exercises improve students' conceptual understanding of core probability ideas, and whether these exercises add value beyond traditional handwritten practice. Simulation-based inference has become a mainstream approach in introductory statistics and data science courses, where students specify a model, run many simulated trials, and study the resulting distributions to build intuition (Rossman and Chance 2014). By contrast, probability courses are still typically taught through lectures, algebraic derivations, and pen-and-paper problem sets, with relatively little systematic integration of coding and simulation into the core learning experience (Sheikh 2024). We have developed a reusable library of simulation-based coding exercises and accompanying lesson notes for several core units in an undergraduate probability sequence, designed to be dropped into existing courses with minimal modification. We then implemented a four-arm, full factorial randomized classroom experiment that crossed two dimensions: the presence versus absence of coding-based exercises, and the presence versus absence of additional handwritten-style exercises, yielding a control group, a coding-only group, a handwritten-only group, and a coding+handwritten group. Using assessments with both multiple-choice and short-answer questions, we compared learning gains across these conditions data science students. Our project provides quantitative evidence on (i) whether coding-based simulation helps students learn probability, and whether its effects are distinct from simply providing more practice, and (ii) whether combining coding with handwritten work yields incremental benefits over either modality alone, along with a set of reusable coding activities and instructor-facing lesson notes that illustrate how to integrate simulation into probability instruction.

Website: https://sainthades.github.io/Simulation-coding-exercises-for-teaching-probability-theory/
Code (Experiment): https://github.com/SaintHades/Simulation-coding-exercises-for-teaching-probability-theory

Code (Teaching Materials): https://github.com/peterbchi/codingprobsim25

1 Introduction 3
2 Related Work 4
3 Methods 6
4 Results 11
5 Discussion 17
6 Conclusion 26
Appendices A1

1 Introduction

Probability is a core part of modern STEM education and everyday decision-making, yet many people find it confusing and unintuitive. Students often struggle with ideas like conditional probability, rare events, and long-run behavior: they may be able to plug numbers into formulas, but still feel unsure about what those formulas mean in real situations. This matters because probabilistic reasoning shows up in many important contexts, such as interpreting medical tests, evaluating risks, or judging the strength of data-driven claims. Traditional probability courses tend to emphasize symbolic derivations and hand-written problem sets, which build algebraic skills but do not always help students develop a clear mental picture of random processes.

In contrast, statistics and data science courses have increasingly adopted simulation-based, coding-centered approaches, where learners write short programs, generate many samples from a model, and use plots and summaries to "see" theoretical ideas in action (Rossman and Chance 2014). Recent work argues that probability teaching has not yet fully taken advantage of these ideas and tools, and that more systematic, application-based use of simulation could make abstract concepts more accessible (Sheikh 2024; Ross and Sun 2019). This emerging literature motivates the need for careful, classroom-based evaluation of how coding and simulation actually affect students' conceptual understanding of probability, beyond anecdotal enthusiasm or small, topic-specific case studies.

Our project directly addresses this gap. We designed simulation-based coding exercises for several key probability topics and embedded them in a four-condition, full factorial experiment that compared a control group, a coding-only group, a handwritten-only group, and a coding+handwritten group. Within this design, we asked two related questions: (i) whether access to coding activities, in any form, leads to additional learning gains beyond traditional materials alone, and (ii) whether combining coding with handwritten work yields benefits beyond using either modality by itself.

Our primary hypothesis was that any condition involving coding will produce larger gains on conceptual understanding than conditions without coding, because simulations make abstract ideas more concrete and visible. Our secondary hypothesis was that the coding and handwritten condition will show an additional gain, as learners both explore stochastic behavior through code and consolidate their understanding through slower, worked symbolic solutions. By measuring learning gains across these conditions and packaging our activities into reusable materials, we aim both to answer a concrete question—whether and how coding actually helps students understand probability better—and to leave behind practical resources that instructors and students can continue to use. Given the centrality of probability in later coursework and in high-stakes real-world decisions, even modest improvements in instruction have the potential to yield meaningful downstream benefits, making this a timely and important topic for systematic study.

2 Related Work

Historically, a college level class in probability theory utilizes pencil and paper techniques as its sole medium. Recently, there has been a push to incorporate technology into probability theory classes, particularly the usage of computer simulation. However, the exact methods for doing so are still not yet well formed. We aim to do that here.

Research within college settings has been done, each with the intention of determining whether incorporating simulation and application-based learning material improves the performance of students on exams. One notable example is the development of the UCLA web-based set of educational material called the Statistics Online Computational Resource (SOCR). This website includes a number of statistical tools and computational widgets ranging from practice problems to computer-generated visualizations of real-life processes. In a study conducted across five different statistics courses, separated into three treatment and two control classes, the effects of SOCR-based curriculum proved to significantly increase the performance of students on both exams and homework (Christou, Dinov and Sanchez 2008). Additionally, the study aimed to draw connections between individual students' learning styles and academic success, also discovering that some learning styles were significant predictors of results in addition to SOCR vs. non-SOCR teaching.

In a similar setting, a study at Purdue University Northwest analyzed the efficacy of simulation-based learning in the context of electrical and computing engineering (ECE) curriculum. This study tested the effects of simulation by incorporating MATLAB into the course and utilized an experimental design that separated two quarters of students into a control and treatment group, keeping the difficulty of the class as consistent as possible for each iteration of the course. MATLAB itself provides a computing environment with visualization and coding tools, and students in the treatment group were given interactive MATLAB simulations for topics spanning signal processing, controls, and more rather than focusing on traditional abstract theory. Students in the treatment group displayed over a 10% increase in mean exam scores (Sheikh 2024) in comparison to the control group. Furthermore, more than 88% of students who participated in the treatment group supported the notion of implementing MATLAB into the course itself, showing enormously positive support for simulation-based learning. This study supports the potential benefits and enhanced knowledge retention that result from using simulation and how it may extend to undergraduate statistics education.

Additionally, research has been done on a Python simulation package called Symbulate (Ross and Sun 2019), which allows students to use the "language of probability" to run simulations without needing to know how the code works. The creators believed that implementing simulations from scratch would not be relevant to the content of the course and therefore not necessary for students to learn. The Symbulate programs are capable of running simulations and creating visualizations. They acknowledge that R or python skills can be useful for getting jobs, but they don't believe that it is the purpose of a probability course. 81% of students in a course that used Symbulate agreed that "The course provided evidence that simulation provides an effective tool for analyzing probability models."

The limitations of these controlled experiments lie in the assignment of groups and small

student population sizes as limitations on need for coding knowledge. Both studies separated groups based on individual courses, meaning that entire classrooms were either designated to be control or treatment groups. This was acknowledged as a key constraint in both reports, as the experimental design opened possibilities for confounding factors such as student composition or semester-specific effects that may have altered the results found. The last study on Symbulate acknowledged that R or python skills can be useful for getting jobs, but they didn't believe it to be the purpose of a probability course.

We aim to address these gaps by implementing a factorial design in our experiment. This entails splitting up our participants into separate groups, with each group receiving a different combination of treatment factors, allowing us to address each possible combination of stimulus and analyze the effects of each in relation to one another. With this, we hope to potentially reduce the effects of confounding variables, increasing the robustness of our results. Some supplementary data we plan to collect includes information on participants including year, major, familiarity with topics, and related courses completed. Additionally, we plan to include a post-experiment assessment that takes after the questions used in the paper from (Sheikh 2024) that can be seen below.

No.	Survey question
1	The MATLAB® simulations improved my understanding and comprehension of abstract probabilistic concepts.
2	The MATLAB® simulations helped me in developing probabilistic intuition.
3	The MATLAB® simulations motivated me to study the subject of probability.
4	The MATLAB® simulations motivated me to study communications, networking, and signal processing; areas of electrical engineering which employ probabilistic analysis.
5	The MATLAB® simulations improved my critical thinking abilities.
6	The MATLAB® simulations improved my ability to solve problems employing probabilistic modeling and analysis.
7	The MATLAB® simulations developed my ability to apply probabilistic concepts to related but different problems in electrical engineering.
8	The MATLAB® simulations helped me with applying the abstract theoretical knowledge in practical contexts.
9	Without MATLAB® simulations, I would not have grasped an in-depth understanding of the probabilistic concepts in electrical engineering.
10	The MATLAB® simulations improved my understanding of probabilistic modeling and performance analysis of various electrical engineering problems in communications, networking, signal processing, and controls.
11	MATLAB® simulations based on real-world engineering applications such as, maximum likelihood signal detection, minimum mean squared error quantization, optimum network buffer allocation, and Kalman filtering, helped foster my interest and motivation in the area of electrical engineering.
12	I would like to have a MATLAB® based lab for this course.

A major difference in our experimental design resides in our measurement of the impact of simulation based exercises on a final assessment. We aim to analyze not only the difference between results from students given only handwritten exercises as opposed to simulation coding exercises but also if doing both handwritten and coding exercises leads to further improvement of scores. To do so we will fit the results data to two linear models to see the effect of handwritten exercises, coding simulation exercises, and both exercises on final

assessment scores. More on our analysis will be done in later sections. In comparison, prior studies focused on aggregate exam and homework scores as metrics in determining success of the treatment. Our statistical approach allows us to examine individual student responses to simulation-based intervention and identify patterns in learning that may get lost in the the granularity of entire sample sizes. Through these methods we aim to produce a more robust understanding of how simulation can influence teaching probability theory.

3 Methods

3.1 Data Collection

We recruited participants from Data Science, Math, and Economics classes at UCSD. Data collection took place during the first half of winter quarter. Selection of these courses aimed to target students with comparable quantitative backgrounds, ensuring a similar baseline level of mathematics and coding skills for completing the tasks in our study. We conducted participant outreach through class announcements and emails, distributed through the professors of each respective course. Students who chose to participate were first asked to complete an intake form administered through Google Forms. Background information such as name, email address, data science affiliation, math affiliation, class standing, related coursework, and sentiment regarding familiarity with statistics, Python, and Chebyshev's inequality were collected from each participant.

3.2 Experimental Design

Our project generated its own dataset through a controlled 2×2 full-factorial experimental design. The study manipulated two factors related to learning activities: (i) whether students completed coding exercises, and (ii) whether they completed handwritten exercises. Each factor has two levels ("done' vs. "not done'), producing four experimental conditions:

- No coding, no handwritten
- Coding only
- Handwritten only
- Both coding and handwritten

Participants were randomly assigned to one of these four conditions to ensure comparability across groups. To meaningfully measure the effect of different teaching methods, we gave the students exercises and supplementary material based on their assigned experimental group. Materials were administered through Gradescope, and each student received an email detailing which tasks to complete based on their assigned grouping. All materials focused on the topic of Chebyshev's Inequality, a concept which the majority of participants (approximately 60%) reported not being very familiar with. After completing their assigned learning activities, every student completed the same final assessment, producing a quantitative score that served as the primary outcome variable to evaluate differences in

learning outcomes across each combination of experimental conditions.

3.3 Analysis of Overall Assessment Patterns

Before conducting formal statistical analysis, it is useful to outline the general patterns that we expect to observe in the assessment outcomes based on the instructional design of the experiment.

The four experimental conditions differ in the extent to which students engage with structured learning activities. Section 1 serves as the control condition, where students complete neither coding exercises nor additional handwritten practice. Sections 2 and 3 introduce a single form of structured engagement, while Section 4 combines both coding and handwritten exercises.

Based on prior literature on simulation-based learning and practice-based reinforcement, we expect that sections involving structured engagement activities may exhibit stronger overall performance compared with the control group. In particular, students who complete both coding and handwritten exercises may benefit from the complementary nature of these activities: coding exercises allow students to explore probabilistic behavior through simulation, while handwritten exercises may help consolidate conceptual understanding through more deliberate symbolic reasoning.

From this perspective, we may anticipate a general ordering of outcomes in which Section 4 tends to perform strongest, followed by Sections 2 and 3, with Section 1 potentially exhibiting lower or more variable performance. However, these expectations represent qualitative patterns rather than formal statistical conclusions. The purpose of the subsequent statistical analysis is to evaluate whether any observed differences in assessment outcomes are statistically significant.

3.4 Statistical Analysis

We estimate differences in assessment scores across the four experimental groups using a linear regression model:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 x_2),$$

where x_1 is the indicator for completing coding exercises, x_2 is the indicator for completing handwritten exercises, and the interaction term $x_1 x_2$ captures whether the combination of the two activities produces an additional effect beyond the separate contributions of each.

Two central hypotheses guide the analysis:

1. $H_0 : \beta_1 = \beta_2$, assessing whether coding exercises and handwritten exercises have equal effects on learning outcomes.
2. $H_0 : \beta_3 = 0$, testing whether the two activities interact.

To evaluate the first hypothesis, we will compare a model that includes separate coefficients for coding and handwritten exercises against a nested model that constrains these two

effects to be equal. This will be done using an ANOVA model comparison. To evaluate the interaction hypothesis, we will examine the significance of the interaction coefficient in the full model.

3.5 Power Analysis and Implications for Sample Size

Because this is a factorial experiment, sample size requirements depend on the expected effect sizes for each cell of the design and the variability of assessment scores. In our simulation framework, each "replicate" corresponds to one complete 2×2 block of students: four students assigned respectively to the four treatment combinations. Thus, using 25 replicates corresponds to 25 students per treatment cell and a total sample size of 100 students across the entire experiment. The probability of detecting main effects or the interaction naturally increases with the number of students per cell. To evaluate whether the proposed sample size is adequate, we conduct simulation-based power analyses under different assumed effect structures, mirroring the linear model used in our primary analysis.

Power calculations were performed by generating simulated datasets with 25 students per cell (100 students in total) and repeating this process 1,000 times for each assumed set of means. For each simulated dataset we fit the model, test the relevant hypotheses, and compute the proportion of times the null hypotheses are correctly rejected. These simulations provide direct empirical estimates of power for both the main effects and the interaction and allow us to assess whether our planned sample size is sufficient for the intended analyses.

Power for Main Effects. To evaluate the power to detect differences between the two learning activities (coding vs. handwritten), we varied the group mean for the $(x_1 = 1, x_2 = 0)$ condition while holding the other three group means fixed at (3, 3.5, μ_{10}, 6) with standard deviations set to 1. With 25 participants per cell and 1,000 Monte Carlo iterations per scenario, the estimated power increases smoothly as the main-effect contrast grows. The resulting power curve is summarized in Table 1. When μ_{10} is close to 3.5, effect sizes are small and power remains below 0.10. Power increases rapidly for larger deviations, exceeding 0.80 by approximately $\mu_{10} = 4.3$ and approaching 1.0 for $\mu_{10} \geq 4.6$. These results indicate that the experiment is adequately powered to detect moderate-to-large main effects but may have limited sensitivity to very small differences in learning outcomes.

Table 1: Estimated power for detecting main effects as μ_{10} varies (replicates = 25, iter = 1000).

μ_{10}	Power
3.0	0.241
3.1	0.187
3.2	0.106
3.3	0.074
3.4	0.032
3.5	0.036
3.6	0.039
3.7	0.069
3.8	0.139
3.9	0.250
4.0	0.376
4.1	0.486
4.2	0.620
4.3	0.771
4.4	0.888
4.5	0.934
4.6	0.977
4.7	0.991
4.8	0.997
4.9	1.000
5.0	1.000

Power for the Interaction. To assess the ability to detect the interaction term x_1x_2, we fixed the first three group means at (3, 3.5, 4.5) and varied the fourth cell mean μ_{11} from 3 to 7 in steps of 0.2. When $\mu_{11} = 5$, the design is exactly additive, implying no interaction; in this case, the simulated power correctly returns to the nominal Type I error rate (approximately 0.05). As shown in Table 2, power rises sharply as μ_{11} deviates from 5. A deviation of roughly ±1.0 yields power around 0.70, and deviations of about ±1.4 or larger produce power exceeding 0.90. These results indicate that the proposed design can reliably detect moderate or strong interactions but is not sensitive to very small departures from additivity.

Implications. Taken together, these simulations show that with 25 participants per cell, the study is well-powered to detect moderate-to-large main effects and interactions, but may struggle to detect subtle effects that correspond to small differences across learning conditions. The power analyses therefore both justify the chosen sample size for the Capstone study and provide context for interpreting null results: a non-significant effect may reflect either the absence of a meaningful difference or the presence of a very small effect that the design is not powered to detect.

Table 2: Estimated power for detecting the interaction as μ_{11} varies (replicates = 25, iter = 1000).

μ_{11}	Power
3.0	1.000
3.2	0.996
3.4	0.978
3.6	0.927
3.8	0.859
4.0	0.714
4.2	0.511
4.4	0.315
4.6	0.159
4.8	0.056
5.0	0.047
5.2	0.077
5.4	0.171
5.6	0.327
5.8	0.512
6.0	0.715
6.2	0.852
6.4	0.918
6.6	0.976
6.8	0.998
7.0	0.999

3.6 How Analyses Will Be Communicated

The results of the study will be communicated through multiple complementary forms:

- **Regression summaries**: tables reporting estimates, standard errors, confidence intervals, and p-values.
- **Factor-level visualizations**: plots showing mean assessment scores for the four treatment combinations.
- **ANOVA model comparisons**: highlighting evidence for or against equal effects of the two activities.
- **Interaction diagnostics**: visualizing whether the effects of coding and handwritten exercises reinforce or substitute for one another.
- **Power analysis results**: summaries of simulated power curves demonstrating the adequacy of the chosen sample size.

3.7 Summary

In summary, our project generates a structured experimental dataset using a full-factorial design, and analyzes the resulting assessment outcomes using linear modeling, hypothesis testing, and simulation-based power evaluation. All analytical results will be communicated through regression tables, visualizations, and clearly organized summaries suitable for inclusion in the final Capstone report.

4 Results

4.1 Exploratory Overview of Assessment Outcomes

Before presenting the results of the formal statistical models, we first examine the overall distribution of assessment outcomes across the four instructional groups. This exploratory analysis provides a descriptive overview of patterns in the data and helps contextualize the statistical tests reported in the following subsections.

Figure 1 presents the distribution of final assessment scores across the four instructional groups.

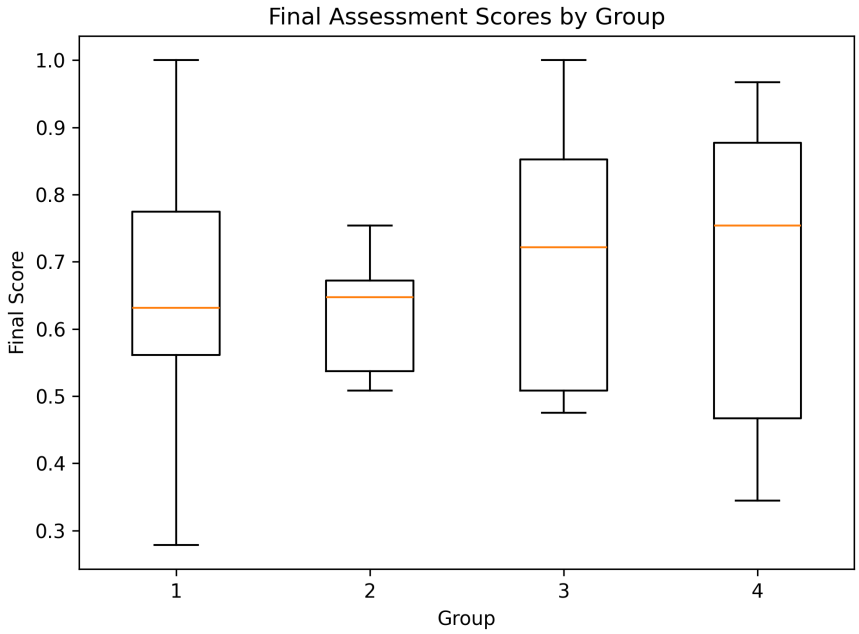

Figure 1: Final Assessment Scores by Group.

Overall, moderate differences can be observed between groups. Sections involving structured engagement activities (Sections 2–4) tend to exhibit slightly higher median scores compared with Section 1, where students did not complete either coding or handwritten exercises.

Among the structured engagement groups, Section 4 (both coding and handwritten work) appears to show relatively strong performance in the upper portion of the score distribution. In contrast, Section 1 exhibits a wider spread of scores, suggesting greater heterogeneity in learning outcomes.

To further account for differences in students' reported confidence, we also examine a confidence-adjusted score metric. Figure 2 shows the distribution of adjusted final scores by section.

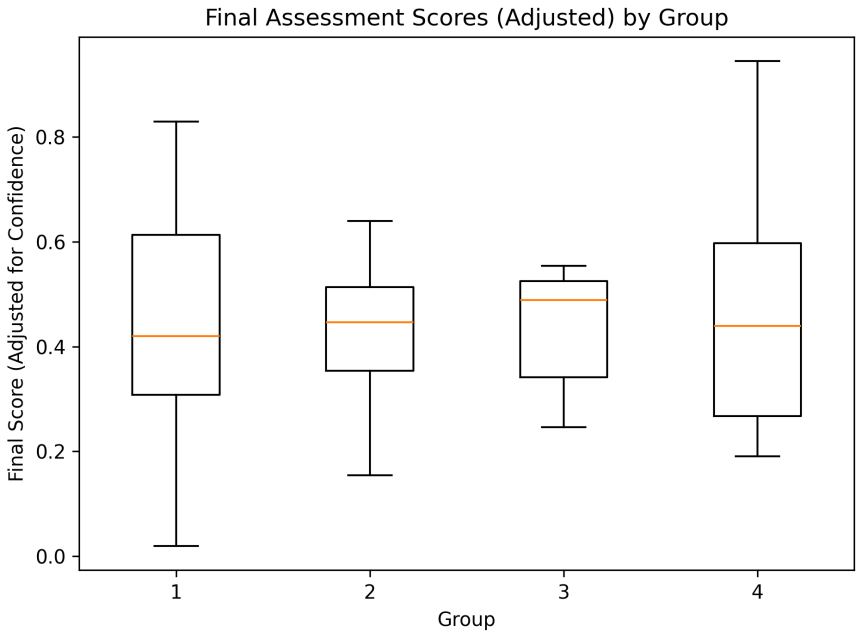

Figure 2: Final Assessment Scores (Adjusted) by Group.

Across all groups, adjusted scores are slightly lower than the corresponding raw final scores. This reduction reflects instances where incorrect answers were reported with relatively high confidence. The decrease appears somewhat more pronounced for Section 1, suggesting that students without structured engagement activities may exhibit weaker alignment between perceived confidence and actual correctness.

Taken together, Figures 1 and 2 suggest modest differences across instructional conditions. In particular, sections involving structured engagement activities appear to show somewhat stronger performance and more consistent outcomes. However, exploratory analysis alone cannot determine whether these observed patterns are statistically meaningful.

To formally evaluate these differences, we now turn to the statistical models and hypothesis tests introduced in the Methods section.

4.2 First hypothesis result for final score

The exploratory analysis presented earlier suggested that sections involving structured engagement activities (Sections 2–4) tend to exhibit slightly stronger performance than the

control group. In particular, the EDA plots indicated that Section 4 (both coding and handwritten exercises) appears to perform somewhat better in the upper range of the score distribution. To determine whether these observed patterns reflect statistically meaningful differences rather than visual variation in the data, we now formally evaluate the hypotheses using regression models and ANOVA model comparisons.

The following presents the results of an ANOVA test comparing two models used to predict the final score. The first two figures display the coefficient statistics for each model. The null hypothesis is that the coefficient for coding is equal to the coefficient for handwritten exercises ($H_0 : \beta_1 = \beta_2$). An ANOVA test is used to compare the two models, and the results of this test are shown in the third figure.

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.629326	0.048367	13.011567	5.027419e-17	0.531969	0.726683
coding	0.001821	0.076474	0.023818	9.811005e-01	-0.152114	0.155756
handwritten	0.070128	0.083774	0.837108	4.068598e-01	-0.098500	0.238755
coding:handwritten	-0.017221	0.119812	-0.143736	8.863365e-01	-0.258391	0.223948

Figure 3: First model: $\beta_0 + \beta_1(\text{coding}) + \beta_2(\text{handwritten}) + \beta_3(\text{interaction})$

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.629326	0.048145	13.071521	2.889696e-17	0.532471	0.726181
I(coding + handwritten)	0.031095	0.065610	0.473942	6.377356e-01	-0.100895	0.163086
coding:handwritten	-0.007463	0.118567	-0.062947	9.500758e-01	-0.245989	0.231062

Figure 4: Second model: $\beta_0 + \beta_1(\text{coding} + \text{handwritten}) + \beta_3(\text{interaction})$

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	47.0	1.960964	0.0	NaN	NaN	NaN
1	46.0	1.936968	1.0	0.023995	0.569846	0.454168

Figure 5: ANOVA test result for final score

With p-value equal to 0.453168, significantly greater than standard significant level 0.05. Which tells us that we fail to reject the assumption that coding exercises and handwritten exercises have equal effects on learning outcomes.

Although the exploratory plots suggested modest differences in score distributions across instructional sections, the statistical test indicates that these differences are not large enough to be considered statistically significant within the current sample. In other words, while structured engagement activities appeared visually associated with slightly higher scores, the evidence does not support a statistically detectable difference between the effects of coding exercises and handwritten exercises.

4.3 Second hypothesis result for final score

Another pattern suggested by the exploratory analysis is that the section combining both coding and handwritten exercises (Section 4) appeared to exhibit somewhat stronger performance in the upper portion of the score distribution. This observation raises the question of whether the combination of the two activities produces an additional benefit beyond their individual effects. To evaluate this possibility, we test whether the interaction term in the regression model is statistically significant.

Refer back to model: $\beta_0 + \beta_1(\text{coding}) + \beta_2(\text{handwritten}) + \beta_3(\text{interaction})$, the below figure show us the result for the second hypothesis to see whether $\beta_3 = 0$.

	0	1	2	3
0	Model:	OLS	Adj. R-squared:	-0.041
1	Dependent Variable:	final_score	AIC:	-12.6511
2	Date:	2026-03-08 00:32	BIC:	-5.0030
3	No. Observations:	50	Log-Likelihood:	10.326
4	Df Model:	3	F-statistic:	0.3621
5	Df Residuals:	46	Prob (F-statistic):	0.781
6	R-squared:	0.023	Scale:	0.042108

Figure 6: Model for predicting final score

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.629326	0.048367	13.011567	5.027419e-17	0.531969	0.726683
coding	0.001821	0.076474	0.023818	9.811005e-01	-0.152114	0.155756
handwritten	0.070128	0.083774	0.837108	4.068598e-01	-0.098500	0.238755
coding:handwritten	-0.017221	0.119812	-0.143736	8.863365e-01	-0.258391	0.223948

Figure 7: Coefficients for predicting final score

The p-value for the coefficient of the interaction term (coding*handwritten) is approximately 0.8863, which is much greater than the significance level of 0.05. Therefore, we fail to reject the null hypothesis that $\beta_3 = 0$. This suggests that there is no statistically significant evidence that the interaction term contributes to predicting the final score.

Thus, although Section 4 appeared to perform somewhat better in the exploratory visualizations, the regression results suggest that this apparent advantage may be attributable to natural variation in the data rather than a systematic interaction effect between coding and handwritten exercises.

4.4 First hypothesis result for adjusted final score

Because the exploratory analysis also examined confidence-adjusted scores, we repeat the same statistical procedures using the adjusted final score as the outcome variable. This al-

lows us to assess whether accounting for students' confidence levels changes the conclusions suggested by the raw score analysis.

The process and steps are the same as in the previous part, but we are now predicting the adjusted final score. The following presents the results of an ANOVA test comparing two models for predicting the adjusted final score. The first two figures display the coefficient statistics for each model. The null hypothesis is that the coefficient for coding is equal to the coefficient for handwritten exercises ($H_0 : \beta_1 = \beta_2$). An ANOVA test is used to compare the two models, and the results of this test are shown in the third figure.

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.434426	0.049614	8.756188	2.330031e-11	0.334559	0.534293
coding	-0.012295	0.078446	-0.156733	8.761412e-01	-0.170199	0.145609
handwritten	0.027140	0.085933	0.315829	7.535607e-01	-0.145835	0.200115
coding:handwritten	0.019879	0.122901	0.161749	8.722122e-01	-0.227508	0.267267

Figure 8: First model: $\beta_0 + \beta_1(\text{coding}) + \beta_2(\text{handwritten}) + \beta_3(\text{interaction})$

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.434426	0.049179	8.833537	1.496645e-11	0.335490	0.533362
I(coding + handwritten)	0.004606	0.067020	0.068722	9.455021e-01	-0.130221	0.139433
coding:handwritten	0.025513	0.121114	0.210651	8.340706e-01	-0.218137	0.269163

Figure 9: Second model: $\beta_0 + \beta_1(\text{coding} + \text{handwritten}) + \beta_3(\text{interaction})$

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	47.0	2.046130	0.0	NaN	NaN	NaN
1	46.0	2.038132	1.0	0.007998	0.18051	0.672916

Figure 10: ANOVA test result for adjusted final score

For the adjusted final score, the p-value from the ANOVA test comparing the two models is 0.672916, which is larger than the p-value obtained for the unadjusted final score (0.453168). Since the p value is greater than significant level 0.05, we fail to reject the null hypothesis that coding and handwritten exercises have equal effects on learning outcomes. Comparing the result for adjusted final score to the result of unajusted final score, the larger p-value (0.672916 > 0.453168) provides even stronger evidence in favor of retaining the null hypothesis.

This result is also consistent with the exploratory plots, where the adjusted scores showed similar overall patterns to the raw scores but with slightly lower values across all sections. The statistical results confirm that these descriptive differences remain statistically indistinguishable under the current model.

4.5 Second hypothesis result for adjusted final score

Again, refer back to model: $\beta_0 + \beta_1(\text{coding}) + \beta_2(\text{handwritten}) + \beta_3(\text{interaction})$, the below figure show us the result for the second hypothesis to see whether $\beta_3 = 0$.

	0	1	2	3
0	Model:	OLS	Adj. R-squared:	-0.056
1	Dependent Variable:	final_score_adj	AIC:	-10.1056
2	Date:	2026-03-08 00:33	BIC:	-2.4575
3	No. Observations:	50	Log-Likelihood:	9.0528
4	Df Model:	3	F-statistic:	0.1288
5	Df Residuals:	46	Prob (F-statistic):	0.943
6	R-squared:	0.008	Scale:	0.044307

Figure 11: Model for predicting adjusted final score

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.434426	0.049614	8.756188	2.330031e-11	0.334559	0.534293
coding	-0.012295	0.078446	-0.156733	8.761412e-01	-0.170199	0.145609
handwritten	0.027140	0.085933	0.315829	7.535607e-01	-0.145835	0.200115
coding:handwritten	0.019879	0.122901	0.161749	8.722122e-01	-0.227508	0.267267

Figure 12: Coefficients for predicting adjusted final score

The p-value for the coefficient of the interaction term (coding*handwritten) is approximately 0.8722, which is very close to the p-value(0.8863) for the unadjusted final score. Both values are much greater than the significance level of 0.05. Therefore, we fail to reject the null hypothesis that $\beta_3 = 0$. This suggests that there is no statistically significant evidence that the interaction term contributes to predicting the adjusted final score.

Taken together, the statistical results indicate that while the exploratory visualizations suggested modest differences across instructional conditions, these differences do not translate into statistically significant effects in the regression models. This outcome may reflect the relatively small sample size of the experiment, which limits the power to detect subtle differences across learning conditions.

4.6 Summary of Statistical Results

Taken together, the statistical analyses presented above provide a consistent picture of the relationship between instructional conditions and assessment outcomes.

First, the ANOVA model comparisons for both the raw final score and the confidence-adjusted final score fail to reject the null hypothesis that coding exercises and handwritten exercises have equal effects on learning outcomes ($H_0 : \beta_1 = \beta_2$). Although the exploratory visualizations suggested that sections involving structured engagement activities tended

to exhibit slightly stronger performance, the regression results indicate that the difference between coding-based and handwritten exercises is not statistically significant within the current dataset.

Second, the hypothesis tests for the interaction term also fail to reject the null hypothesis ($H_0 : \beta_3 = 0$) for both outcome variables. This indicates that there is no statistically significant evidence that combining coding and handwritten exercises produces an additional effect beyond the individual contributions of each activity.

Finally, the results obtained using the confidence-adjusted final score closely mirror those obtained using the raw final score. While the adjusted scores are slightly lower across all sections, reflecting cases where incorrect answers were reported with high confidence, the overall statistical conclusions remain unchanged.

Overall, these findings suggest that although exploratory patterns in the data indicate modest differences across instructional conditions, the formal statistical tests do not detect significant effects of instructional mode under the current experimental design.

5 Discussion

5.1 Subquestion-Level Variability

Beyond the overall assessment outcomes presented earlier, examining performance at the level of individual questions provides additional insight into how students responded to different types of probability problems.

To analyze item-level variation, scores for each subquestion were rescaled to the interval $[0, 1]$, allowing comparisons across questions with different point allocations. Summary statistics were then computed for each subquestion, including mean score and standard deviation.

The resulting distributions reveal substantial heterogeneity in item difficulty and discriminative power. Some subquestions exhibit ceiling effects, where the mean score approaches the upper bound and variability across students is limited. Other items display larger standard deviations or lower mean scores, indicating greater variability in student understanding.

To illustrate these patterns more concretely, we focus on four representative subquestions selected based on extreme statistics in the score distribution: the highest mean score, the lowest mean score, the highest standard deviation, and the lowest standard deviation.

5.2 Subquestions of Interest

Highest Mean Score (Subquestion 2.3).

Subquestion 2.3 achieves the highest mean score among all items. This question asks stu-

dents to interpret a probability statement related to Chebyshev's inequality within a relatively direct computational setting.

The high mean score suggests that most students were able to correctly apply the inequality once the problem structure was clearly specified. Performance on this item appears relatively consistent across sections, indicating that straightforward procedural applications may be less sensitive to differences in instructional mode.

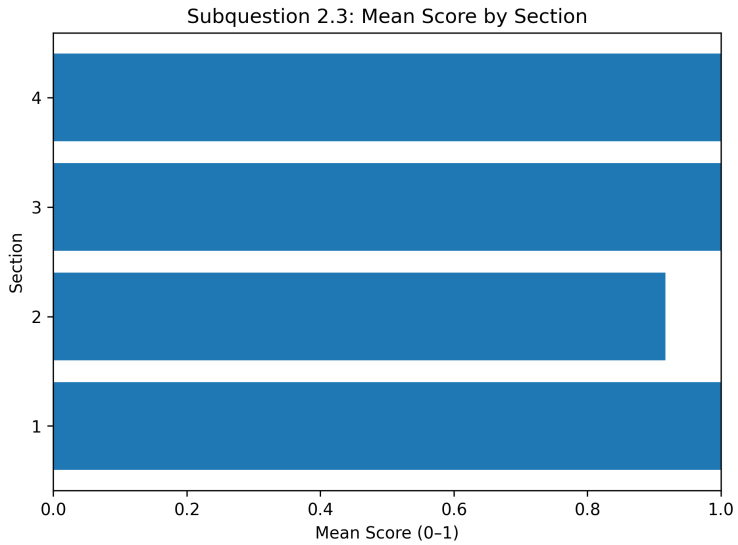

Figure 13: Highest Mean Score (Subquestion 2.3).

Figure 13 shows section-wise mean performance for the highest-mean subquestion.

Lowest Mean Score (Subquestion 5.2).

Subquestion 5.2 exhibits the lowest mean score among the assessed items. This question concerns the conceptual interpretation of the Weak Law of Large Numbers (WLLN) and requires identifying the correct qualitative statement about convergence of sample means.

Compared with computational tasks, conceptual probability statements may require a deeper understanding of asymptotic reasoning. This additional conceptual complexity may explain the lower overall performance on this item.

Figure 14 reports section-wise mean performance for the lowest-mean subquestion.

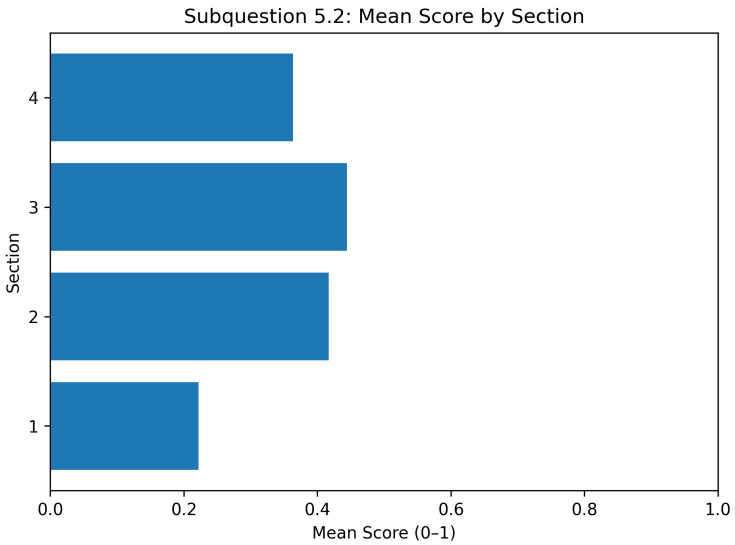

Figure 14: Lowest Mean Score (Subquestion 5.2).

Highest Variability (Subquestion 3.4).

Subquestion 3.4 shows the largest standard deviation among all items, indicating substantial variation in student responses.

This question requires applying Chebyshev's inequality within a multi-step reasoning process involving both algebraic manipulation and interpretation of probabilistic bounds. Tasks that combine conceptual understanding with symbolic reasoning may amplify differences in preparation and problem-solving strategies across students.

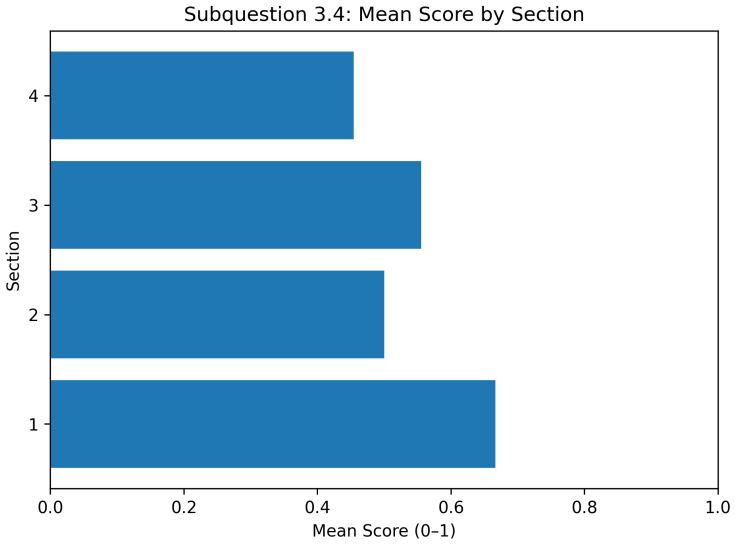

Figure 15: Highest Variability (Subquestion 3.4).

Figure 15 shows section-wise mean performance for the subquestion with the largest standard deviation.

Lowest Variability (Subquestion 1.1).

Subquestion 1.1 has the smallest standard deviation across items. This question focuses on a basic conceptual property of Chebyshev's inequality and appears to have been well understood by most students.

The relatively small spread of scores indicates that performance on this item is highly consistent across students. This suggests that the underlying concept was sufficiently covered across all instructional modes, resulting in minimal differentiation between sections.

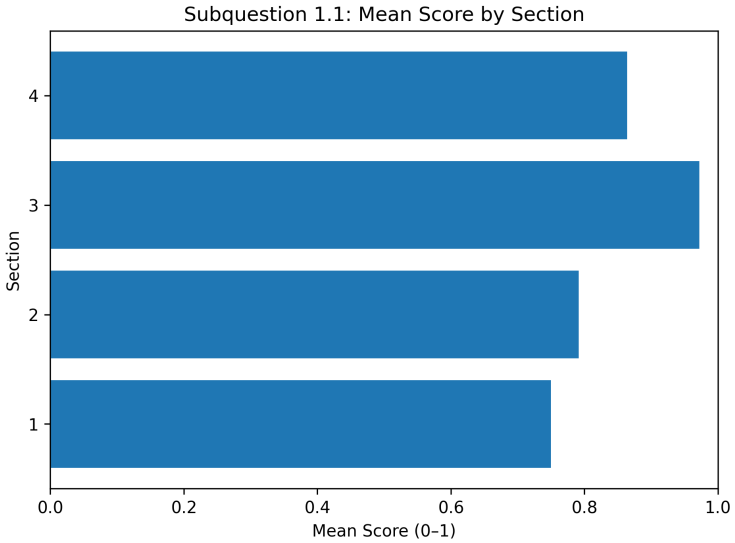

Figure 16: Lowest Variability (Subquestion 1.1).

Figure 16 shows section-wise mean performance for the subquestion with the smallest standard deviation.

5.3 Confidence Calibration

Student confidence levels provide an additional perspective on metacognition by examining how well reported confidence aligns with correctness.

Figure 17 shows the relationship between mean score and reported confidence level aggregated across all students. The results reveal a clear positive association: higher confidence levels correspond to higher average correctness. This pattern suggests that student confidence is generally informative about answer correctness, indicating reasonable overall calibration.

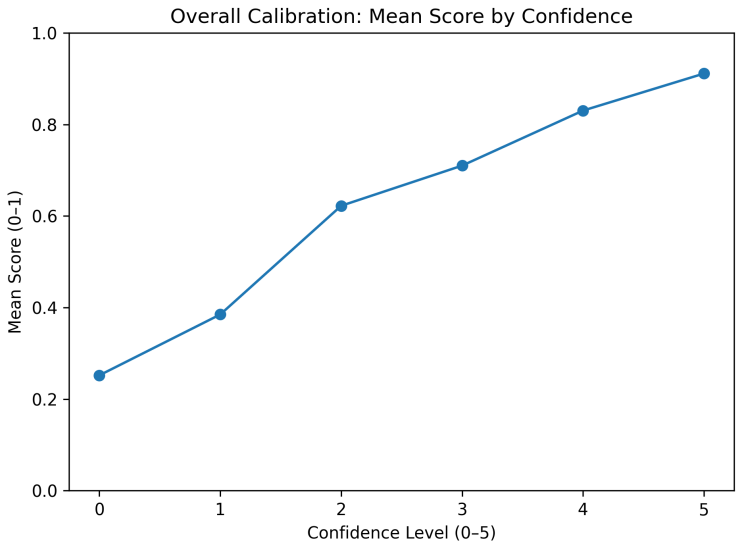

Figure 17: Overall Calibration: Mean Score by Confidence.

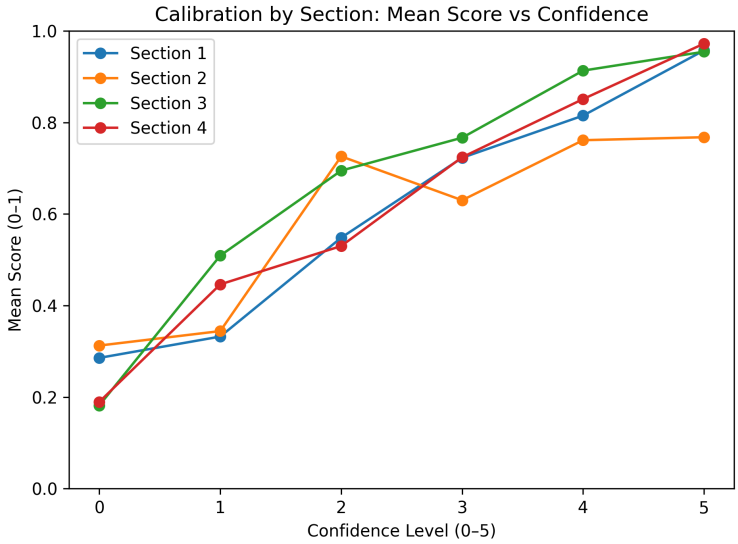

Figure 18: Calibration by Section: Mean Score vs Confidence.

Figure 18 further examines calibration separately for each instructional group. All sections display a broadly increasing relationship between confidence and score, indicating that the overall calibration pattern holds across instructional conditions.

However, the strength and smoothness of this relationship vary somewhat across sections. Some groups exhibit more consistent monotonic trends, whereas others display fluctuations at intermediate confidence levels. These variations may reflect differences in how students interpret uncertainty or evaluate their own understanding under different engagement modes.

5.4 Interpretation of Findings

Taken together, the exploratory analyses highlight several patterns that complement the formal statistical results presented earlier.

First, sections involving structured engagement activities (coding, handwritten work, or both) tend to exhibit slightly stronger and more consistent assessment outcomes compared with the section without such activities. However, the statistical tests reported in the Results section indicate that these differences are not statistically significant within the sample studied.

Second, the subquestion-level analysis reveals substantial heterogeneity in item difficulty and discriminative power. Some questions are answered correctly by most students, while others produce substantial variation in responses, indicating differences in conceptual understanding across topics.

Finally, the confidence calibration analysis suggests that students' reported confidence is generally aligned with correctness, although calibration patterns vary somewhat across instructional groups.

Together, these findings provide descriptive insight into assessment outcomes across instructional conditions and help contextualize the statistical tests reported in the Results section.

5.5 Additional Statistical Analyses

Throughout our experiment, the main limitation we encountered was the small sample size. As a result, the R values and adjusted R^2 values for both the model predicting the final score and the model predicting the adjusted final score were really low, even though the section features theoretically contribute directly to the score.

Moving beyond testing our hypotheses, we decided to improve our model by adding additional features collected from the intake form. The new features are:

- `Source of recruitment` – Categorical
- `DSC affiliation` – Categorical
- `Math affiliation` – Categorical
- `Academic year` – Categorical
- `Statistics courses taken` – Categorical
- `Confidence in statistics skills` – Numerical
- `Familiarity with Chebyshev's inequality` – Numerical
- `Python skill level` – Numerical

The numerical variables are measured on a scale from 1 to 5, and one-hot encoding is applied to the categorical variables. Here is a clarification of how we name each feature (feature name : feature description):

- Features starting with `Heard`: `Source of recruitment`
- Features starting with `DSC`: `DSC affiliation`
- Features starting with `MATH`: `Math affiliation`
- Features starting with `Year`: `Academic year`
- Features starting with `Stats`: `Statistics courses taken`
- `stats_confidence`: `Confidence in statistics skills`
- `chebyshev_familiarity`: `Familiarity with Chebyshev's inequality`
- `python_skill_level`: `Python skill level`

Including a large number of features in the model can lead to overfitting. To address this issue, we applied backward feature selection to remove features that are not important for predicting the score. The backward feature selection stop when the adjusted R^2 value of the model no longer increase.

This section describes the backward feature selection procedure for the model predicting the adjusted final score. At the beginning, the model has an adjusted $R^2 = 0.096272$ with all features included.

1st iteration: In the first iteration of backward feature selection, the features `Heard_4`, `Heard_1`, `Heard_3`, and `Heard_2` were removed, with p-values equal to 0.257, 0.526, 0.618, and 0.882, respectively. Since four features were generated after applying one-hot encoding to the recruitment feature, they were treated as a group during the selection process. Although `Heard_4` has a relatively lower p-value, the other three features have relatively high p-values, so all four features were removed together. After removing these features, the adjusted R^2 increased to 0.168767.

2nd iteration: The second iteration of backward feature selection dropped `MATH_major` and `MATH_minor`, with p-values equal to $0.528, 0.716$. These two features were generated after applying one-hot encoding to the Math affiliation. Both features have relatively high-values. After removing these features, the adjusted R^2 increased to 0.203840.

Following features dropped respectively:

Feature name:	p-value:
`python_skill_level`	0.662
`stats_confidence`	0.449
`chebyshev_familiarity`	0.486

The p-value for `chebyshev_familiarity` was calculated after `stats_confidence` had been removed from the model. After removing these features, the adjusted R^2 increased to 0.222685, then to 0.231890, and finally to 0.242632.

The backward feature selection procedure stops at this point, since removing additional features would decrease the adjusted R^2 value. The following is the final model for predicting the adjusted final score.

	0	1	2	3
0	Model:	OLS	Adj. R-squared:	0.243
1	Dependent Variable:	final_score_adj	AIC:	-18.9976
2	Date:	2026-03-08 00:35	BIC:	7.7707
3	No. Observations:	50	Log-Likelihood:	23.499
4	Df Model:	13	F-statistic:	2.208
5	Df Residuals:	36	Prob (F-statistic):	0.0305
6	R-squared:	0.444	Scale:	0.031767

Figure 19: Final model for predicting adjusted final score

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.424988	0.078878	5.387943	0.000005	0.265017	0.584960
Year_transfer_2	0.234096	0.100089	2.338882	0.025010	0.031106	0.437085
Year_1	0.068492	0.058416	1.172486	0.248702	-0.049981	0.186966
Year_transfer_1	-0.101551	0.072552	-1.399701	0.170162	-0.248692	0.045591
Year_4	0.099106	0.067718	1.463514	0.152004	-0.038232	0.236443
Stats_2	0.162814	0.087471	1.861350	0.070876	-0.014585	0.340214
coding_handwritten	0.040992	0.111120	0.368898	0.714364	-0.184370	0.266353
Stats_1	0.123414	0.113365	1.088644	0.283545	-0.106500	0.353328
handwritten	0.056697	0.078346	0.723677	0.473941	-0.102195	0.215589
Stats_3	0.030567	0.077864	0.392566	0.696955	-0.127349	0.188483
coding	0.010626	0.078625	0.135145	0.893250	-0.148832	0.170084
DSC_minor	-0.219788	0.082015	-2.679862	0.011037	-0.386121	-0.053455
DSC_major	-0.171539	0.078396	-2.188111	0.035234	-0.330534	-0.012545
Year_2	0.004956	0.077406	0.064024	0.949305	-0.152032	0.161943
Year_3	0.119889	0.067525	1.775481	0.084273	-0.017058	0.256837

Figure 20: Coefficients for predicting adjusted final score with intake features

The model indicates that including `Academic Year`, `Statistics Courses Taken`, and `DSC Affiliation` as input features helps better predict the adjusted final score while reducing the risk of overfitting. We observe a substantial improvement in the model by comparing the increase in the adjusted R^2 value from 0.096272 to 0.242632.

Considering the limitations encountered in our experiment, the model also suggests that there may be additional features that contribute to predicting the adjusted final score. These limitations will be discussed in the next subsection.

5.6 Limitations

Heterogeneity of assessment question types

Several limitations of the current study should be considered when interpreting the results.

First, the overall assessment aggregates performance across multiple question types that differ in their cognitive demands. Some items primarily emphasize procedural computation, while others require deeper conceptual reasoning or interpretation of probabilistic statements. Because these question types may favor different learning modalities, their effects may partially offset one another when aggregated into a single overall score.

For example, simulation-based coding exercises may be particularly helpful for developing intuition about stochastic behavior or long-run patterns, whereas handwritten exercises may better support symbolic reasoning and step-by-step derivations. When both types of questions appear within the same assessment, improvements associated with one instructional approach on certain question types may be counterbalanced by advantages of the other approach on different items. As a result, differences between instructional conditions may become less visible in the aggregate score used for the primary statistical tests.

The subquestion-level analyses presented above provide some preliminary evidence for this possibility. Variation in item difficulty and response variability suggests that different questions may capture distinct aspects of probabilistic understanding. This raises the possibility that instructional effects may become more apparent when examining specific categories of problems rather than the overall assessment score.

Future work could therefore benefit from designing assessments that explicitly distinguish between different types of probability tasks (for example, computational problems, conceptual interpretation, and simulation-based reasoning). Analyzing instructional effects within more narrowly defined problem categories may provide a clearer picture of how coding-based and handwritten exercises influence different dimensions of probabilistic understanding.

Heterogeneity in confidence reporting across students

Another limitation arises from the role of student confidence in the adjusted score analysis. While the calibration plots indicate a positive relationship between reported confidence and correctness, confidence levels themselves represent a heterogeneous factor across students.

Students may interpret and use the confidence scale differently. Some students may systematically report higher confidence, while others may adopt more conservative confidence estimates even when their answers are correct. These individual differences introduce additional variability into the confidence-adjusted score metric.

As a result, variation in confidence reporting may partially obscure differences attributable to instructional conditions. Even when instructional approaches influence students' conceptual understanding, heterogeneous confidence behavior across students may dampen observable differences in the adjusted scores.

Future work could address this limitation by examining confidence calibration at a finer granularity, such as normalizing confidence measures across students or incorporating

student-level calibration parameters into the analysis.

6 Conclusion

This study aimed to investigate whether teaching probability and statistical concepts through coding exercises leads to better learning outcomes compared to traditional handwritten exercises. To evaluate this question, we tested two hypotheses: whether coding and handwritten exercises have equal effects on learning outcomes ($H_0 : \beta_1 = \beta_2$), and whether the two instructional approaches interact with each other ($H_0 : \beta_3 = 0$).

Based on the models estimated in this study, we failed to reject both null hypotheses. In other words, the results do not provide strong statistical evidence that coding exercises lead to different learning outcomes compared to handwritten exercises, nor do they suggest a significant interaction between the two activities in affecting student performance.

However, these results should be interpreted with caution. The study faced several limitations, most notably the relatively small sample size and the heterogeneity in both assessment question types and student confidence reporting. These factors may have introduced additional variability into the outcome measures and potentially reduced the statistical power needed to detect instructional effects.

Future work should therefore focus on collecting a larger sample and designing assessments that more clearly distinguish between different types of probability tasks. Such refinements may help reveal whether coding-based instruction provides advantages for specific aspects of probabilistic reasoning, such as simulation-based intuition or conceptual understanding.

References

Christou, Nicolas, Ivo D. Dinov, and Juana Sanchez. 2008. "Pedagogical utilization and assessment of the statistic online computational resource in introductory probability and statistics courses." *Computers Education* 50: 284–300. [Link]

Ross, Kevin, and Dennis L. Sun. 2019. "Symbulate: Simulation in the Language of Probability." *Journal of Statistics Education* 27(1): 12–28. [Link]

Rossman, Allan J., and Beth L. Chance. 2014. "Using Simulation-Based Inference for Learning Introductory Statistics." *Wiley Interdisciplinary Reviews: Computational Statistics* 6(4): 211–221. [Link]

Sheikh, Waseem. 2024. "An Intuitive, Application-Based, Simulation-Driven Approach to Teaching Probability and Random Processes." *International Journal of Electrical Engineering & Education* 61(1): 17–57. [Link]

Appendices

A.1 Contributions A1

A.1 Contributions

- **Marguerite Church:** Created simulation based coding exercises on Cauchy distributions and Exponential distributions. Created the intake form for the experiment and graded the open ended assessment questions.
- **Rihui Ling:** Created simulation based coding exercises on Conditional Probability, Binomial Random Variable, and Geometric Random Variable. Tracking new students who fill out the intake form, adding them to Gradescope. Monitor student process and update the status to Alyssa. Downloaded and organized all data of intake form and grade from Gradescope. Cleaned those data for further analysis.
- **Jingjun Ni:** Created lesson notes, coding problem sets for the Poisson distribution and related topics, relationship between exponential distribution and gamma distribution. Worked on statistics analysis Code.
- **Alyssa Tallada:** Created simulation based coding exercises for joint, marginal, and conditional distributions. Emailed participating students intake information and distributed vouchers to completed students.
- **Yunqi Zhang:** Preparation of the draft of the handwritten problem set on Chebyshev's Inequality and built the EDA code framework and analysis pipeline. Creation of the coding exercises, handwritten problem sets, and assessments on Central Limit Theorem and Hypergeometric Variables.